

TokTickIT Lab 3

Final Submission Evidence - Answer Part 1 through Answer Part 9

Final released main: 55d80fe109eec03e683e394da2a51a1e59ddb5c4

Post-release evidence PR: <https://github.com/Chxtamos/-TokTickIT-/pull/86>

Prepared 2026-10-03

Answer Part 1: Git Use with Engineering Workflow

Lab 3 used feature branches and peer-reviewed PRs into `lab3-staging`, followed by release PR #84 from `lab3-staging` to `main`. PR #84 was approved on corrected head `aa45eb7` and merged as `55d80fe` on 2026-10-02. The implementation/review history is rendered from `docs/lab-03/reviewer.md`; it records reviewer identity, requested changes, correction evidence, approvals, CI and merge commits for the real PR set. Issue #65 is deliberately reopened while post-release evidence PR #86 is under review; the Project item is Started and PR #86 is PR Review. It must become Done only after the final evidence PR is reviewed/merged and the human visual checklist is signed off.

Links: repository <https://github.com/Chxtamos/-TokTickIT-> ; release PR <https://github.com/Chxtamos/-TokTickIT-/pull/84> ; evidence PR <https://github.com/Chxtamos/-TokTickIT-/pull/86> ; Project <https://github.com/users/Chxtamos/projects/6> .

Answer Part 2: Spec DD

The Sprint 3 contract is in `docs/lab-03/specification.md`, `ui-spec.md`, and `api-spec.md`. It defines the Sprint Goal, scope/exclusions, numbered FRs and BRs, the Requester/IT Staff/Administrator authorization matrix, authenticated ownership, migration from Development Requester identity, password/session decisions, Ticket workflow, Public Comments versus Internal Notes, Administrator safety rules, AC-01 through AC-32, and Product Definition of Done. The contract was created and reviewed before the main implementation sequence; PR #68 is the contract review record.

Rendered Spec DD excerpt

Lab 3 Sprint Engineering Specification | Contract version: 1.0, prepared 2026-09-15. Status: prepared for student/peer review; implementation and product completion are not claimed. | Source: supplied `Lab\_3\_sheet.pdf`, all 18 pages, especially sections 4-10 and 14. Lab 2 baseline: local `main` commit `439ee7e`. See README.md for the inspected baseline and implementation-plan.md for the proposed Issues. | The handout is the assignment source. Requirements explicitly fixed by it remain mandatory. Concrete choices below resolve its intentionally unspecified details; they are project decisions, not quotations from the instructor. Sample images do not override written exclusions (including email reset, Service Actions and advanced user management). | 1. Sprint Goal | Extend the existing Requester MVP with email/password authentication, mandatory initial-password change, backend role/ownership enforcement, an operational IT Staff Ticket Queue and Ticket Detail, shared Public Comments, restricted Internal Notes, and minimalist Administrator User Management. Preserve existing Tickets, Attachments, numbering, reference data, and permitted Requester behavior while replacing the Development Requester selector. | 2. Stakeholder Request | Real accounts must replace the temporary requester context. Requesters continue handling their own submissions and communicate with support. IT Staff find, own, prioritize and progress work. Administrators manage accounts. Every screen and API observes the authorization matrix; hidden controls alone never grant protection. All screens extend the existing Zen Green application. | 3. Scope | Included: the sprint goal; forward-only Prisma/PostgreSQL migrations and idempotent local seed; all four normative contract documents; traceable tests; Issues, feature PRs, peer review, staged integration, visual evidence and the final nine-part PDF. | Excluded: self-registration; invitations or password-reset email; MFA, social login and SSO; Actions Taken/Service Actions; SLA/escalation/notifications; analytics beyond simple queue totals; organizations/departments/multi-tenancy; multiple roles per user; user deletion, bulk operations, import/export, role/account-history screens, profile pictures, account unlocking or advanced identity management; mandatory admin pagination or multi-column sorting; production deployment/cloud changes. Requester editing/deleting saved Ticket fields is not added: Lab 2 implemented read-only detail and Attachment management. | 4. Functional Requirements | - **FR-01:** Authenticate an active, provisioned User using email/password with safe failure feedback and bounded attempts. | - **FR-02:** A successful initial-password login permits only current-user retrieval, mandatory password change and logout before normal application access; its restricted session expires exactly 15 minutes after issuance regardless of activity. | - **FR-03:** Change password after current-password verification and valid new-password confirmation; atomically record passwordChangedAt, increment User.version/updatedAt, revoke old sessions and persist a fresh normal session before its cookie is sent. | - **FR-04:** Restore the session through current-user retrieval; display name/role; logout invalidates server access and clears account-specific UI state. | - **FR-05:** Enforce the matrix in section 5 on the backend and reflect it in navigation, controls and direct screen guards. | - **FR

Answer Part 3: Test DD and Traceability

`docs/lab-03/tests.md` is the source-of-truth test plan and AC-to-test index. Exact final `main` `55d80fe` evidence: Client CI #134 / run 37054900319 passed 15 files and 93 tests plus production build; Server CI #147 / run 37054900447 applied all four migrations, ran PostgreSQL integration, passed 30 files and 219 tests plus server build; E2E CI #116 / run 37054900427 ran migrations, seed twice, isolated credential provisioning, and passed 20/20 Playwright tests. Post-release evidence head `8bd84be` passes Client #137, Server #150, and E2E #119 with 25/25 Playwright tests including the final responsive capture suite.

Rendered Test DD final-main excerpt

Final main and Issue #65 completion evidence - 2026-10-03

Release PR #84 merged `lab3-staging` into `main` as merge commit `55d80fe109eec03e683e394da2a51a1e59ddb5c4` on 2026-10-02. The exact final-main push reran all hosted workflows with zero required skips: Client CI run `37054900319` (#134) passed 15 files / 93 tests and the production build; Server CI run `37054900447` (#147) applied all four migrations, ran the guarded PostgreSQL suites, passed 30 files / 219 tests, and passed the server build; E2E CI run `37054900427` (#116) applied migrations, ran the seed twice, provisioned isolated credentials, and passed 20/20 Playwright tests.

Post-release evidence PR #86 starts from that exact final-main commit and adds only final evidence plus responsive fixes discovered by the required audit. Its exact evidence head `8bd84be` has Client CI `37116872973` (#137) success, Server CI `37116872950` (#150) success, and E2E CI `37116872958` (#119) success with 25/25 Playwright tests. The five added evidence cases cover 1440, 768, 390, 360 CSS-pixel layouts and a 200%-zoom-equivalent 720 CSS-pixel layout, capturing Login, Requester My Tickets, Staff Queue, Staff Ticket Detail, and Administrator User Management. The audit found real horizontal overflow at the 768 Requester breakpoint and 360 Administrator cards; PR #86 corrects those defects rather than hiding the evidence.

Screenshot evidence is stored under `artifacts/lab-03/screenshots/{1440,768,390,360,zoom-200}/`. Automated assertions prove no document-level horizontal overflow for every captured major screen. The screenshots are suitable for the required human visual checklist, but automated capture/inspection is not represented as a human sign-off. A student/reviewer visual sign-off remains the final non-automatable action before Product DoD can be marked complete.

RELEASE-01 automated portion: PASS on exact final `main` `55d80fe` for migration, twice-run seed, unit/API/PostgreSQL/UI/auth/regression/E2E/build evidence, and PASS on PR #86 for the required responsive evidence suite after the audit fixes. VIS-01/A11Y-01 human sign-off remains explicitly pending until a person reviews the stored screenshots and checklist; no AI-authored approval is substituted.

Answer Part 4: AI Use with Reflection

`docs/lab-03/ai-use.md` records the AI workflow and ten selected real prompts. Earlier implementation used the Codex desktop coding agent where the exact model identifier was not exposed; the final evidence pass also used ChatGPT GPT-5.6 Sol. The student's original `My Reflection` remains in that file unchanged in substance. It explains that AI was useful for reading the lab sheet, organizing requirements/specification/API/UI/test planning, while Human-in-the-loop and peer review were necessary because early requirements and generated interpretations needed verification and correction against the Lab 3 handout.

My Reflection (student-authored text)

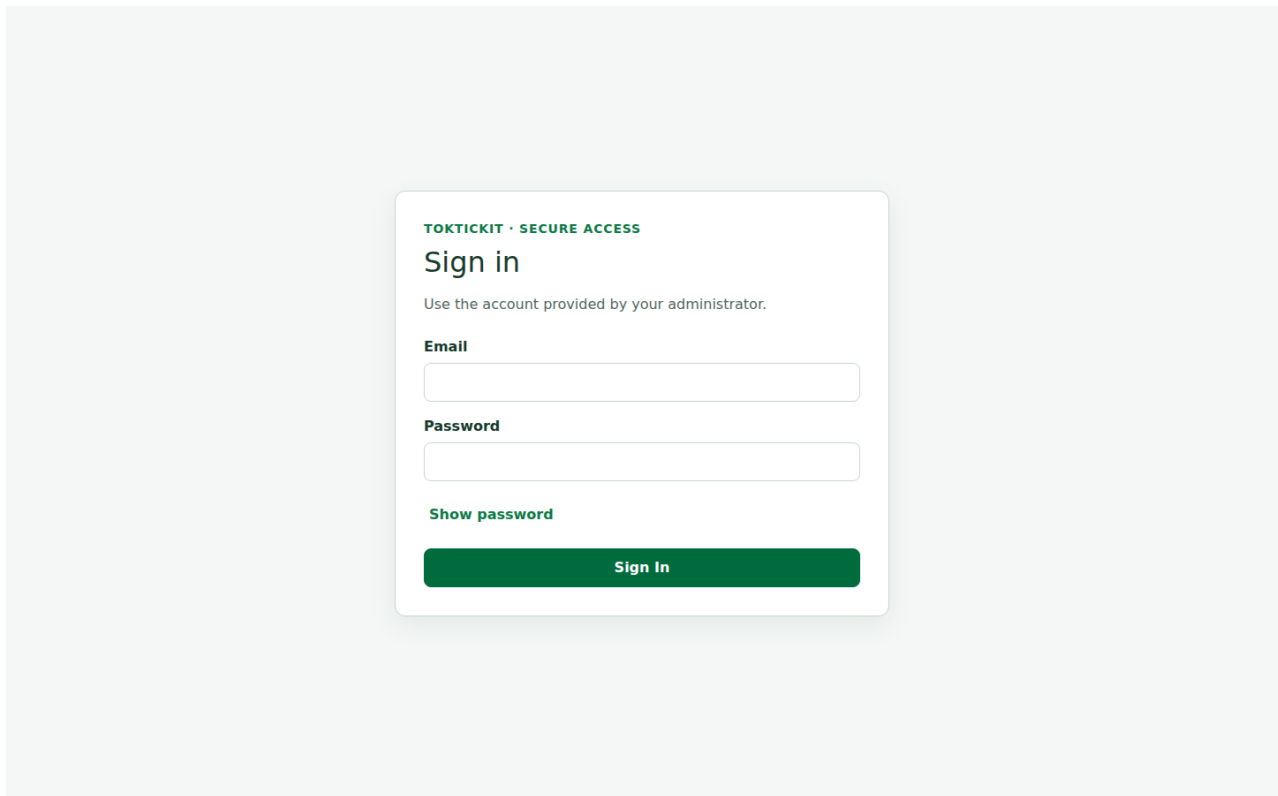
Lab 3 AI Agent labsheet requirements specification, API, UI test plan

Human-in-the-loop peer review Lab 3 labsheet

Answer Part 5: Working Login and Password Change UI

Authentication evidence covers valid/invalid credentials, inactive accounts, mandatory initial-password change, authenticated user/role display, logout and post-logout denial. API tests also cover restricted-session and normal-session expiry boundaries, CSRF/session rotation and safe failures. The screenshot set includes the Login screen at desktop, tablet, mobile and zoom-equivalent layouts. E2E-01 is covered by ``client/e2e/lab-03/authentication.spec.ts``.

1440px evidence



390px evidence

TOKTICKIT • SECURE ACCESS

Sign in

Use the account provided by your administrator.

Email

Password

Show password

Sign In

Answer Part 6: Working IT Staff Ticket Queue UI

The Staff Queue supports realistic data, search, filters, sorting, pagination, ownership/status/priority presentation, open-detail navigation, no-results/failure states and responsive card/table layouts. The final evidence suite captures the queue at 1440, 768, 390 and 360 widths plus the 200%-zoom-equivalent layout. Queue E2E behavior is covered by `client/e2e/lab-03/staff-queue.spec.ts`.

1440px evidence

TokTickITTicket QueueE2E Staff - IT StaffChange PasswordLogout

SHARED IT WORKSPACE
Ticket Queue
Shared work for IT Staff and Administrators. Filters apply across every Ticket.

Search and filters

Ticket Number/Summary search

Ticket number or summary

Category

All Categories

Related System

All Systems

Requested Priority

All Requested Priorities

IT Priority

All IT Priorities

Status

All Statuses

Owner

All

Sort and page size

Sort by

Last updated

Direction

Descending

Page size

10

Clear Filters

49 matching Tickets

Ticket Number	Summary	Category	Requested Priority	IT Priority	Status	Owner	Open
TKT-2026-900011	Seeded Lab 3 Ticket 11 Requester: Arnan Schouk	Software	HIGH	MEDIUM	IN PROGRESS	Unassigned	Open
TKT-2026-900049	Feature 22 visual evidence 1791023772178-89ba3540 Requester: Arnan Schouk	Account and Access	HIGH	HIGH	NEW	Unassigned	Open
TKT-2026-900048	Feature 22 visual evidence 1791023770835-0ba58345 Requester: Arnan Schouk	Account and Access	HIGH	HIGH	NEW	Unassigned	Open
TKT-2026-900047	Feature 22 visual evidence 1791023769471-7a76cb33 Requester: Arnan Schouk	Account and Access	HIGH	HIGH	NEW	Unassigned	Open
TKT-2026-900046	Feature 22 success 1791023765326 Requester: Arnan Schouk	Account and Access	HIGH	HIGH	NEW	Unassigned	Open
TKT-2026-900045	Feature 22 visual evidence 1791023764011-1a5a2088 Requester: Arnan Schouk	Account and Access	HIGH	HIGH	NEW	Unassigned	Open
TKT-2026-900044	Feature 21 responsive ticket 1791023762670-d1db3048 Requester: Arnan Schouk	Account and Access	HIGH	HIGH	NEW	Unassigned	Open
TKT-2026-900043	Feature 21 responsive ticket 1791023761553-afba2748f Requester: Arnan Schouk	Account and Access	HIGH	HIGH	NEW	Unassigned	Open
TKT-2026-900042	Feature 21 responsive ticket 1791023760055-4da45b7f Requester: Arnan Schouk	Account and Access	HIGH	HIGH	NEW	Unassigned	Open
TKT-2026-900041	Feature 20 Attachment lifecycle 1791023758651-c308371b Requester: Arnan Schouk	Account and Access	HIGH	HIGH	NEW	Unassigned	Open

Previous

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Next

390px evidence

[illegible]

Answer Part 7: Working IT Staff Ticket Detail UI

Staff Ticket Detail demonstrates claim/reassign ownership, IT Priority, permitted status changes, Public Comments, private Internal Notes, Attachment continuity, validation/conflict handling and role restrictions. Direct authorization tests prove Requesters cannot retrieve Internal Notes and protected resources do not leak existence.

`client/e2e/lab-03/staff-ticket-flow.spec.ts` and server Lab 3 authorization/workflow suites provide automated evidence; final screenshots capture Operational Controls and the visibly distinct Internal Notes area.

1440px evidence

The screenshot displays the 'TokTickIT' Staff Ticket Detail page for ticket TKT-2026-900011. The interface is organized into several sections:

- Submitted Ticket Information:** Displays ticket details such as Ticket Number (TKT-2026-900011), Ticket Date (Oct 3, 2026, 10:23 AM), Category (Software), Requester (Alan Schulz (alan.schulz@company.com)), Requested Priority (High), and Requested Status (In Progress).
- Operational Controls:** Contains two main sections:
 - Ticket Owner:** Shows the current owner (Shane) and a dropdown to select a new owner, with buttons for 'Assign Owner' and 'Reassign Owner'.
 - IT Priority:** Shows the current priority (High) and a dropdown to select a new priority, with a 'Save IT Priority' button.
- Status Workflow:** Shows the current status (In Progress) and a dropdown to select a new status, with a 'Change Status' button.
- Resolution and Requester Indication:** Shows the resolution status (Not resolved) and the requester's status (Not resolved).
- Attachments:** Displays a list of attachments, including a file named 'staff-ticket.pdf'.
- Public Comments:** A section for public comments, with a 'Post Public Comment' button.
- Internal Notes:** A section for internal notes, with a 'Post Internal Note' button.

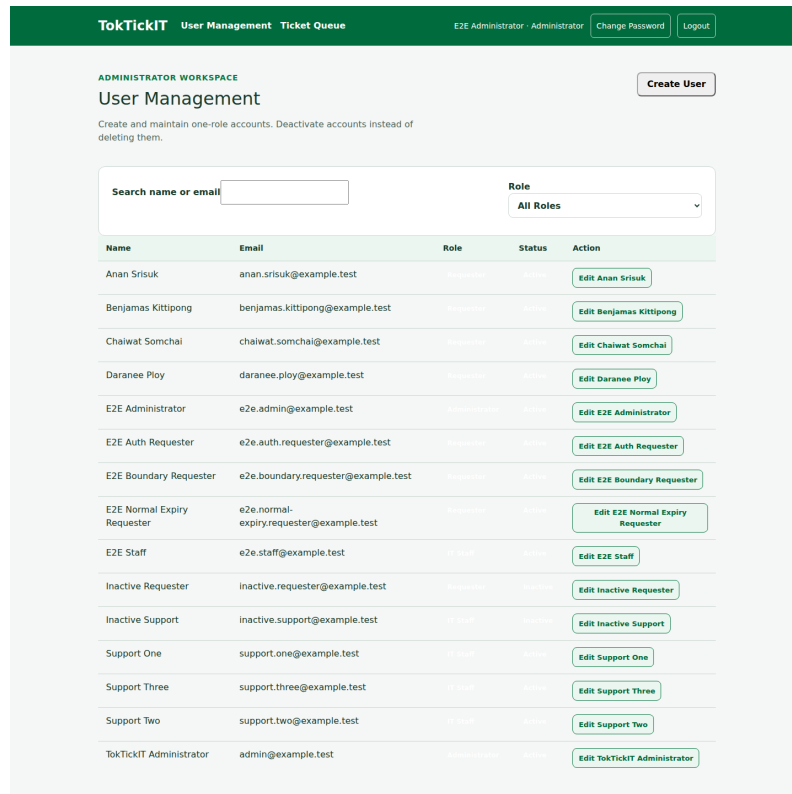
390px evidence

[illegible]

Answer Part 8: Working Administrator User Management UI

User Management provides Name/Email/Role/Status/Edit, search, role filter, create, edit, activation/deactivation and new initial-password behavior. Tests cover duplicate email/invalid input, one-role assignment, self-deactivation prevention, last-active-Administrator protection, session revocation and forbidden access for non-Administrators. `client/e2e/lab-03/admin-user-management.spec.ts` provides E2E-04 evidence. The 360px audit exposed a real overflow defect in Administrator cards; PR #86 fixes it and the rerun passes.

1440px evidence



TokTickIT User Management Ticket Queue E2E Administrator - Administrator [Change Password](#) [Logout](#)

ADMINISTRATOR WORKSPACE [Create User](#)

User Management
Create and maintain one-role accounts. Deactivate accounts instead of deleting them.

Search name or email Role **All Roles**

Name	Email	Role	Status	Action
Anan Srisuk	anan.srisuk@example.test	Administrator	Active	Edit Anan Srisuk
Benjamas Kittipong	benjamas.kittipong@example.test	Administrator	Active	Edit Benjamas Kittipong
Chaiwat Somchai	chaiwat.somchai@example.test	Administrator	Active	Edit Chaiwat Somchai
Daranee Ploy	daranee.ploy@example.test	Administrator	Active	Edit Daranee Ploy
E2E Administrator	e2e.admin@example.test	Administrator	Active	Edit E2E Administrator
E2E Auth Requester	e2e.auth.requester@example.test	Requester	Active	Edit E2E Auth Requester
E2E Boundary Requester	e2e.boundary.requester@example.test	Requester	Active	Edit E2E Boundary Requester
E2E Normal Expiry Requester	e2e.normal-expiry.requester@example.test	Requester	Active	Edit E2E Normal Expiry Requester
E2E Staff	e2e.staff@example.test	Requester	Active	Edit E2E Staff
Inactive Requester	inactive.requester@example.test	Requester	Inactive	Edit Inactive Requester
Inactive Support	inactive.support@example.test	Support	Inactive	Edit Inactive Support
Support One	support.one@example.test	Support	Active	Edit Support One
Support Three	support.three@example.test	Support	Active	Edit Support Three
Support Two	support.two@example.test	Support	Active	Edit Support Two
TokTickIT Administrator	admin@example.test	Administrator	Active	Edit TokTickIT Administrator

390px evidence

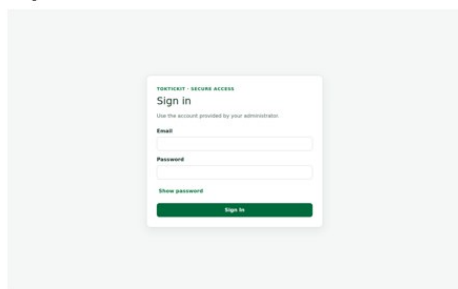
Answer Part 9: Zen Green UI and Responsive Evidence

`docs/lab-03/ui-spec.md` defines the Zen Green visual/accessibility contract. PR #86 adds deterministic evidence for 1440, 768, 390 and 360 CSS-pixel widths and a 200%-zoom-equivalent 720 CSS-pixel layout. It captures Login, Requester My Tickets, Staff Queue, Staff Ticket Detail and Administrator User Management and asserts no document-level horizontal overflow. The first audit exposed two real defects (Requester at 768 and Administrator cards at 360); both were corrected before the green evidence run. Screenshot contact sheets are included in the PDF and the original PNGs are stored under `artifacts/lab-03/screenshots/`.

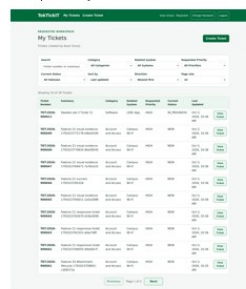
Important completion note: the automated evidence and screenshots are complete, but the handout/contract requires a human visual/accessibility checklist. The coding agent does not fabricate a human sign-off. A student or peer reviewer must inspect the stored screenshots for design consistency, role navigation, badges, editable/read-only styling, validation placement, focus, clipping, overlap, horizontal overflow and absence of secret/Internal Note leakage, then record that sign-off before Issue #65/Product DoD is finally closed.

Major-screen contact sheet - 1440px

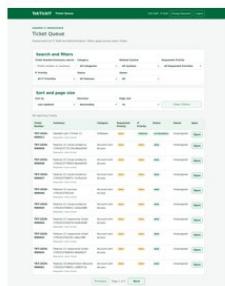
01-login



02-requester-my-tickets



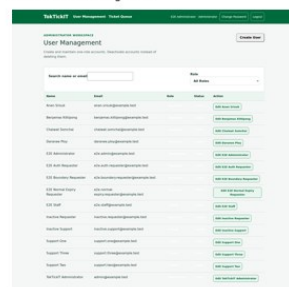
03-staff-queue



04-staff-ticket-detail

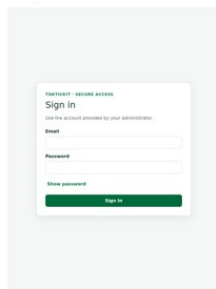


05-admin-user-management



Major-screen contact sheet - 768px

01-login



02-requester-my-tickets



03-staff-queue



04-staff-ticket-detail

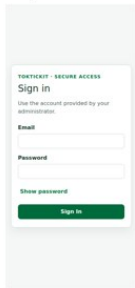


05-admin-user-management



Major-screen contact sheet - 390px

01-login



02-requester-my-tickets



03-staff-queue



04-staff-ticket-detail

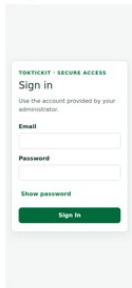


05-admin-user-management



Major-screen contact sheet - 360px

01-login



02-requester-my-tickets



03-staff-queue



04-staff-ticket-detail

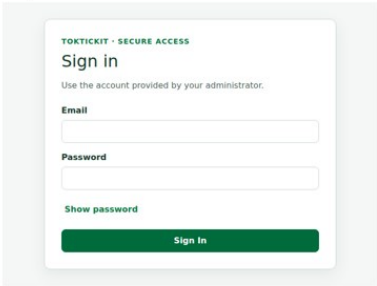


05-admin-user-management



Major-screen contact sheet - 200% zoom equivalent (720 CSS px)

01-login



02-requester-my-tickets



03-staff-queue



04-staff-ticket-detail



05-admin-user-management



Final completion boundary

Automated final-main verification and responsive screenshot generation are complete. The course contract explicitly requires a human visual/accessibility inspection. This PDF does not fabricate that human sign-off. Issue #65 must remain open until a student or peer reviewer inspects the stored screenshots/checklist and PR #86 is reviewed/merged.